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CE 181304

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B.Tech. 3rd Semester End-Term Examination

Civil Engineering

BUILDING CONSTRUCTION AND PLANNING

Bina Chowdhury Central Library
Girijananda Chowdhury University
Hatkhowapara, Azara, Ghy-17

Full Marks – 70

Time – Three hours

The figures in the margin indicate full marks for the questions.

Answer question No. 1 and any *four* from the rest.

1. Fill in the blanks of the following/MCQ: 1 × 10
- (i) Raft foundation are generally preferred when area required for individual footing is more than _____ % of the total area.
 - (ii) In ordinary residential and public buildings the damp proof course is generally provided at _____ level.
 - (iii) The 9 cm × 9 cm side of a brick as seen in the wall face, is generally known as _____.
 - (iv) A stair should not have pitch more than _____ degrees.
 - (v) The member which is placed horizontally to support common rafter of a slopping roof is called _____.
 - (vi) Terrazo flooring is a specially prepared concrete surface containing white cement and _____.
 - (vii) _____ refers to the effect produced by deriving the maximum benefits from the minimum dimensions off a room.
 - (viii) Which direction is suitable for kitchen?
 - (a) S (b) S-E
 - (c) S-W (d) W
 - (ix) As per Assam Unified Building Construction Byelaws, 2022, the minimum front setback of a building from a street is _____ m.
 - (x) LEED stands for _____

[Turn over

2. (a) Enumerate various groups in which buildings are divided. Explain the functional requirements of a building. 8
(b) List different types of foundations used in civil engineering structures. State and explain primary functions of foundation. 7
 3. (a) With a neat sketch define various important parts of a roof truss. 8
(b) Write important points those are to be considered while locating doors and windows of a system. 7
 4. (a) Plan a RCC Dog legged stair case for a residential building with neat and clean sketch showing the required dimensions in which the vertical distance between each floor is 3.36 m and roof is 12 cm in thickness. The size of the stair hall is limited to 4.5 m $\times$ 3 m. 8
(b) What is closer in brick masonry? Explain different types of closers with neat figures. 7
 5. (a) What are the essential components of floor? State and explain salient features of different types of flooring. 8
(b) Write important requirements of good plaster. Explain different defects in Plastering. 7
 6. (a) A bungalow is to be constructed on a plot size 30m $\times$ 50m with G+1 storeys. Permissible F.S.I. is 0.8 and front margin is 4 m and side margins are 3 m each. If maximum construction is to be made on the ground floor, compute the area that can be constructed on first floor. 8
(b) What is a Green Building? What are the different rating system for assessing green buildings in India. Discuss any one of them. 7
 7. (a) What are the general fire safety requirements for buildings? As per NBC of India how buildings are classified on the basis of fire resistance. 8
(b) Explain the factors that need to be considered in building planning. 7
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